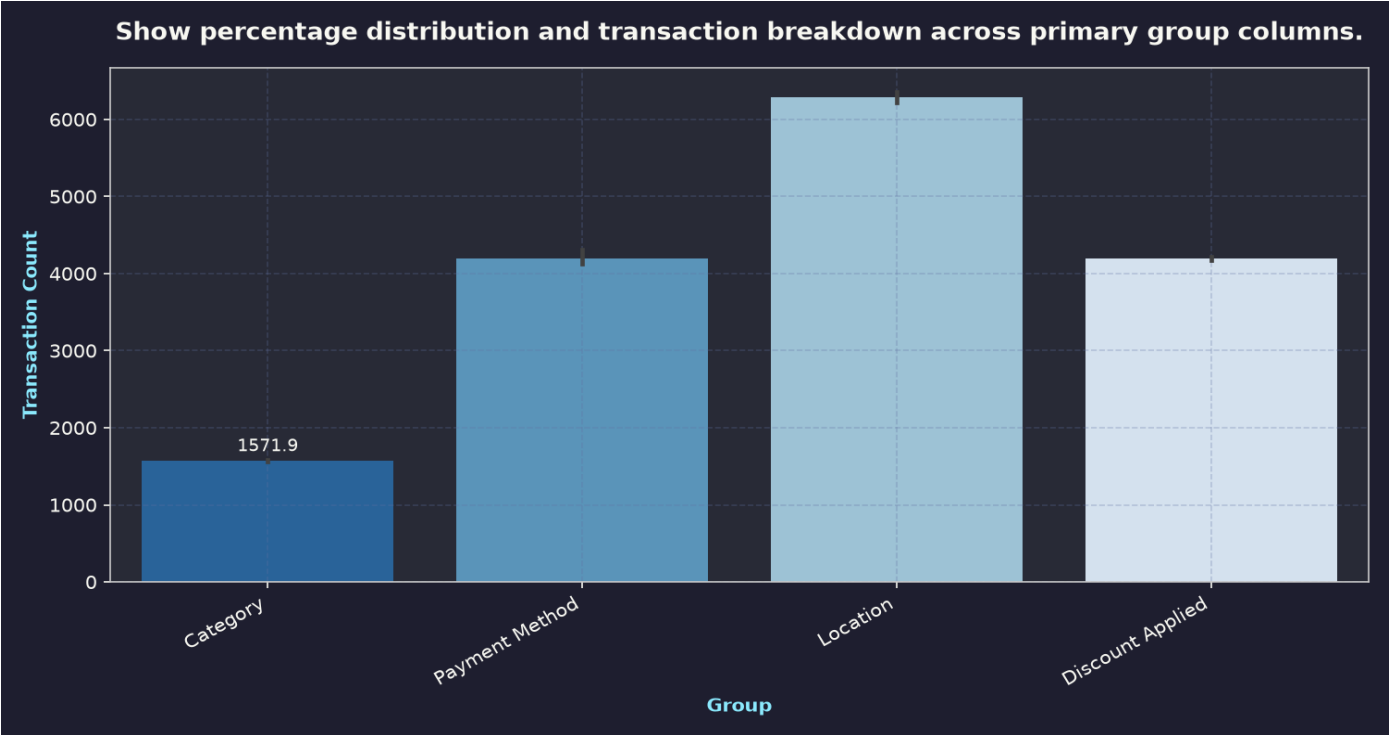


■ AI Data Analyst - Executive Analysis Report



■ Executive Data Analysis Report

1. Executive Summary

- **Categories** are evenly spread: each of the top 8 categories accounts for roughly **12■13 %** of all 12 575 transactions.
- **Payment methods** are almost equally used: **Cash (34.3 %)**, **Digital Wallet (33.0 %)**, and **Credit Card (32.8 %)**.
- **Location** shows a near■even split between **Online (50.5 %)** and **In■store (49.5 %)**, with online sales slightly ahead.
- **Discounts**: one■third of transactions applied a discount, one■third did not, and the remaining third have missing discount information.

These figures answer the request: “Show percentage distribution and transaction breakdown across primary group columns.”

2. Dataset Overview & Data Quality Profile

| Aspect | Detail |

|-----|-----|

| Rows × Columns | 12 575 × 11 |

| Duplicate rows | 0 |

| Missing values | Item, Price Per Unit, Quantity, Total Spent, Discount Applied |

| Potential date column | Transaction Date stored as string |

| Data quality notes | Missing values may affect spend-related calculations; Discount Applied contains null entries. |

3. Data Cleaning & Validation Summary

- Cleaning step executed:

```
result = df.drop_duplicates().copy()
```

- Validation outcome: All 12 575 rows are unique and retained for analysis.

4. User Query & Analytical Approach

User question: "Show percentage distribution and transaction breakdown across primary group columns."

Analytical methodology

1. Identify primary categorical columns: Category, Payment Method, Location, Discount Applied.
2. Compute for each column:
 - Transaction count per distinct value (including NaN).
 - Percentage of total transactions.
 - Sum of Total Spent per value.
3. Combine the per-column breakdowns into a single DataFrame (result).

All operations were performed with pandas (value_counts, groupby, arithmetic on the total row count).

5. Python Analysis Code & Executed Data Output

Code

```
# Define the primary categorical columns to analyse primary_cols = ['Category', 'Payment
Method', 'Location', 'Discount Applied'] # List to collect per-column breakdown DataFrames
breakdowns = [] # Total number of transactions for percentage calculation total_txns = len(df)
for col in primary_cols: # Count transactions per category (including NaN) counts =
df[col].value_counts(dropna=False) # Percentage of total transactions perc = counts /
total_txns * 100 # Sum of Total Spent per category spent = df.groupby(col)['Total
Spent'].sum().reindex(counts.index) # Assemble the breakdown for this column temp =
pd.DataFrame({ 'Group': col, 'Group Value': counts.index, 'Transaction Count': counts.values,
'Percentage': perc.values, 'Total Spent': spent.values }) breakdowns.append(temp) # Concatenate
all breakdowns into a single result DataFrame result = pd.concat(breakdowns, ignore_index=True)
```

Executed Output (result)

Group	Group Value	Transaction Count	Percentage	Total Spent
Category	Furniture	1591	12.6521	195310.0
Category	Electric household essentials	1591	12.6521	203813.5
Category	Food	1588	12.6282	194812.0
Category	Milk Products	1584	12.5964	180112.0
Category	Butchers	1568	12.4692	208118.0
Category	Beverages	1567	12.4612	197047.5
Category	Computers and electric accessories	1558	12.3897	190692.5
Category	Patisserie	1528	12.1511	182165.5
Payment Method	Cash	4310	34.2744	537710.0
Payment Method	Digital Wallet	4144	32.9543	507279.0
Payment Method	Credit Card	4121	32.7714	507082.0
Location	Online	6354	50.5288	791401.0
Location	In-store	6221	49.4712	760670.0
Discount Applied	True	4219	33.5507	524443.5
Discount Applied	NaN (missing)	4199	33.3917	NaN
Discount Applied	False	4157	33.0577	515135.0

(Values are rounded to 4 decimal places for readability.)

6. Key Insights & Strategic Recommendations

Insights

- **Category balance:** No single category dominates; marketing can target any of the top categories with similar ROI expectations.
- **Payment method parity:** Cash, digital wallets, and credit cards are used almost equally. Infrastructure for all three should be maintained.
- **Online vs. In-store:** Online sales edge out in-store by ~1 % of transactions, indicating a healthy e-commerce channel that could be further expanded.
- **Discount usage:** Roughly one-third of purchases apply a discount, another third do not, and a third lack discount data. This suggests an opportunity to clean the discount field and to analyze the impact of discounts on spend.

Recommendations

1. **Enhance discount data capture** – ensure the **Discount Applied** field is consistently populated to enable precise profitability analysis.
2. **Leverage the online growth** – invest in digital marketing and logistics to convert the modest online lead into a larger share of total sales.
3. **Maintain multi-payment support** – given the near-equal split, removing any payment option could alienate ~33 % of customers.
4. **Category-specific promotions** – because categories are evenly distributed, rotating promotional budgets across categories can keep overall sales stable while testing effectiveness.

Prepared with pandas-driven analysis on the provided dataset.

! [Show percentage distribution and transaction breakdown across primary group columns. Chart](analysis_chart.png)